Lecture 9 // Sept 22 // Singularities in $\mathcal{M}$, part 2

Recall

We saw that connections whose stabilizer is larger than $\pm\mathrm{Id}$ lead to singular points of $\mathcal{M}$, and this does happen.

If E is split and ∇ is split, then $\mathrm{Stab}(\nabla)$ is at least as large as S^1. Precisely, we had the following lemma:

If E is a $k \neq 0$ SU(2) bundle, then $\mathrm{Stab}(\nabla)$ is larger than $\pm I$ iff ∇ is split, in which case $\mathrm{Stab}(\nabla) \cong S^1$.

Towards proving this lemma, we discussed parallel transport and holonomy.

The general question is what are the possibilities of $\mathrm{Stab}(\nabla)$? A priori, we know that $\mathrm{Stab}(\nabla) \subseteq \mathcal{G}$, and this fact actually restricts what $\mathrm{Stab}(\nabla)$ must be as follows.

Lemma 1

$g \in \mathrm{Stab}(\nabla)$ iff $D_\nabla g = 0$, where $D_\nabla : \mathrm{End}(E) \to \mathrm{End}(T^*M \otimes E)$, i.e., g is parallel transported by D_∇ along every γ.

Proof. $D_\nabla g$ is characterized by the Leibniz rule:

$$\nabla(gs) = (D_\nabla g)s + g(\nabla s)$$

After moving terms, we get

$$g^{-1}\nabla(gs) = g^{-1}(D_\nabla g)s + \nabla s$$

By definition, $g^*(\nabla)s = g^{-1}\nabla(gs)$.

Thus

$$\begin{aligned} & g \in \mathrm{Stab}(\nabla) \text{ iff } g^*(\nabla) = \nabla \\ \Leftrightarrow\ & \nabla(s) = g^{-1}(D_\nabla g)s + \nabla s \quad \forall s \\ \Leftrightarrow\ & g^{-1}(D_\nabla g)s = 0 \quad \forall s \\ \Leftrightarrow\ & D_\nabla g = 0 \end{aligned}$$

□

Moreover, we know that the parallel transport is unique. So we can also conclude the following.

Lemma 2

If $g \in \mathrm{Stab}(\nabla)$, then g is determined by its value at a single point.

Remark 3

It's not true that any g can be transported: g need not be globally defined! But we know $g \in \mathrm{Stab}(\nabla)$, so a priori, it is a global object, so parallel transport makes sense.

Equivalently, we can write this as for any p, the restriction map $\mathcal{G} \to \mathrm{Aut}(E_p)$ is injective on $\mathrm{Stab}(\nabla)$,

$$g \longmapsto g_p.$$

Thus (via restriction) $\mathrm{Stab}(\nabla)$ can be considered as a subgroup of $\mathrm{SU}(2)$. In fact, $\mathrm{Stab}(\nabla)$ has a nice connection with the holonomy of ∇.

Recall that for a fixed $p \in M$, the holonomy gives a map

$$\mathrm{hol}_p : \{\text{loops based at } p\} \longrightarrow \mathrm{Aut}(E_p) = \mathrm{SU}(2)$$

$$\gamma \longmapsto \mathrm{hol}_p^{\nabla}(\gamma)$$

Very explicitly, under hol_p, each γ gets sent to a specific linear transformation E_p to E_p, e.g., a matrix transformation of the fibre, e.g., an element of $\mathrm{SU}(2)$. If γ is varied to another γ', we get another matrix transformation.

By considering every γ', the set of corresponding matrices is $\mathrm{im}(\mathrm{hol}_p)$, and since $\mathrm{hol}_p(\gamma_2 \circ \gamma_1) = \mathrm{hol}_p(\gamma_2) \circ \mathrm{hol}_p(\gamma_1)$, $\mathrm{im}(\mathrm{hol}_p)$ forms a subgroup of $\mathrm{SU}(2)$.

Lemma 4
The restriction of $\mathrm{Stab}(\nabla)$ to a point p is the centralizer of $\mathrm{im}\,\mathrm{hol}_p$.

Proof. ($\subseteq$): Take $g \in \mathrm{Stab}(\nabla)$. We want to show that g_p commutes with $\mathrm{hol}_p(\gamma)$ for every loop γ. We know that $\mathrm{hol}_p^{g^*(\nabla)}(\gamma) = g_p^{-1} \circ \mathrm{hol}_p^{\nabla}(\gamma) \circ g_p$.

Since $g \in \mathrm{Stab}(\nabla)$, we know $g^*(\nabla) = \nabla$, so $\mathrm{hol}_p^{g^*(\nabla)}(\gamma) = \mathrm{hol}_p^{\nabla}(\gamma)$.

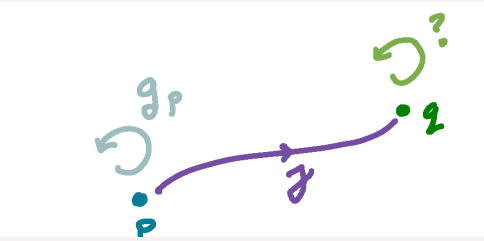
Thus $\mathrm{hol}_p^{\nabla}(\gamma) = g_p^{-1} \circ \mathrm{hol}_p^{\nabla}(\gamma) \circ g_p$ as desired. $\square$

($\supseteq$): Take $g_p \in \mathrm{Aut}(E_p)$. Further assume that $g_p \in Z(\mathrm{im}(\mathrm{hol}_p))$.

We want to show that there exists some $g \in \mathrm{Stab}(\nabla)$ such that the restriction of g to E_p is g_p.

Since $\mathrm{Stab}(\nabla) = \{g \mid D_\nabla g = 0\}$, to define g, we should try to extend g_p via parallel transport by D_∇.

For any other point q, let γ be a path from p to q.



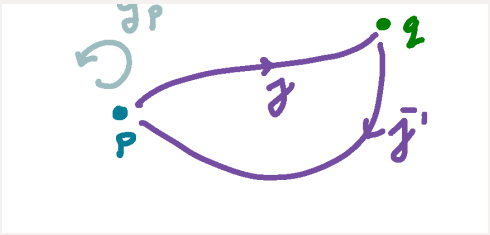
Define g_γ such that

① g_p is extended along γ such that $(D_\nabla)_\gamma g = 0$

② $g_q = \mathrm{hol}_{p,q}(\nabla) \circ g_p \circ \mathrm{hol}_{q,p}(\bar{\gamma})$

We can do this along γ since parallel transport exists. For any path, we can cook up a g_γ such that $D_\nabla(g_\gamma)$ vanishes along γ; this follows by existence of solutions to ODEs.

However a priori, we don't know that this extension along γ gives a well-defined global g. Take γ' another path from p to q, and consider g' defined by γ'.



Then g_q and g'_q are defined as

$$g_q = \mathrm{hol}(\gamma) \circ g_p \circ \mathrm{hol}(\bar{\gamma})$$

$$g'_q = \mathrm{hol}(\gamma') \circ g_p \circ \mathrm{hol}(\bar{\gamma}').$$

If g_q is well defined, $g_q^{-1} = (g'_q)^{-1}$ for any other g'_q, i.e.,

$$g_q (g'_q)^{-1} = \mathrm{id}.$$

Indeed,

$$g_q (g'_q)^{-1} = \mathrm{hol}(\gamma) \circ g_p \circ \mathrm{hol}(\bar{\gamma}) \left(\mathrm{hol}(\gamma') \circ g_p \circ \mathrm{hol}(\bar{\gamma}') \right)^{-1} = \mathrm{id} \qquad \square$$

Let's take stock so far: we know that $\mathrm{Stab}(\nabla)$ is determined by its restriction to a point, which is the centralizer of the holonomy at that point, viewed as a subgroup of $\mathrm{SU}(2)$.

So our question now becomes what are the subgroups of $\mathrm{SU}(2)$?

Fact 5

They are as follows:

Subgroups of SU(2)	Centralizer = Stab(∇)
I	$\mathrm{SU}(2)$
$-I$	$\mathrm{SU}(2)$
S^1 or a conjugate of S^1	S^1
Subgroups of S^1 or conjugate	S^1
Non-abelian	$\pm I$
$\mathrm{SU}(2)$	$\pm I$

This means that the only possibilities for $\mathrm{Stab}(\nabla)$ are $\pm\mathrm{Id}$, S^1, or $\mathrm{SU}(2)$ itself. We are now in the position to prove the lemma.

Lemma 6

If E is a $k \neq 0$ $\mathrm{SU}(2)$ bundle, then $\mathrm{Stab}(\nabla)$ is larger than $\pm I$ iff ∇ is split, in which case $\mathrm{Stab}(\nabla) \cong S^1$.

Proof. We first show that either $\mathrm{Stab}(\nabla) \cong S^1$ or $\pm\mathrm{Id}$. From there, we'll show the former happens iff ∇ is split.

We just saw that there are only three possibilities for $\mathrm{Stab}(\nabla)$, so we need only to rule out $\mathrm{SU}(2)$.

If ∇ is a connection on an $k \neq 0$ bundle, then F_∇ cannot be identically 0 (if it was, $c_2 = k = 0$).

Thus holonomy can't be nontrivial, since the holonomy picks up curvature enclosed by a loop. If this path is varied, hol will vary continuously, so $\mathrm{im}(\mathrm{hol}_p)$ can't be contained in a discrete subgroup. Thus the centralizer cannot be $\pm I$, so the subgroup can't be $\mathrm{SU}(2)$.

Now, suppose $\mathrm{Stab}(\nabla) = S^1$. We just saw that $\mathrm{Stab}(\nabla) = S^1$ implies that at a point $p \in M$, the

element $g_p \in \mathrm{SU}(2)$ must actually be contained in $U(1) \subset \mathrm{SU}(2)$.

Therefore

$$g_p = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix},$$

which induces a splitting of the fibre over p by considering the eigenspaces.

For any other $q \in M$, g_q is obtained from g_p via conjugation with $\mathrm{hol}_{p,q}(\gamma)$, so the eigenspaces of g_q provide a splitting of E_q compatible with that of E_p. Since $\mathrm{Stab}(\nabla)$ is reconstructed from g_p by parallel transport, the splitting of E_p induces a splitting of E.

To Do: Insert proof of other direction. □ □

Having proved this lemma, we can conclude our discussion of why the number of singular points in $\mathcal{M}$ is the number of splittings of E.

We'll now begin a discussion of the other major qualitative feature $\mathcal{M}$: that $\partial\bar{\mathcal{M}} = M$. To do so, we'll first discuss instantons ((A)SD Yang-Mills connections) on S^4, which we'll do by identifying S^4 with $\mathbb{HP}^1$.

Interlude: Review of Quaternions & Some Useful Isomorphisms from Algebra

Definition 7

A quaternion is an expression of the form $q = x + yi + zj + wk$ where $x, y, z, w \in \mathbb{R}$ and $1, i, j, k$ are basis elements such that

$$1 \cdot \alpha = \alpha \text{ for } \alpha = i, j, k, \qquad i^2 = j^2 = k^2 = ijk = -1$$

$$i \cdot j = k, \quad j \cdot k = i, \quad k \cdot i = j, \qquad \text{and} \qquad \alpha \cdot x = x \cdot \alpha \text{ for } \alpha = i, j, k \text{ and } x \in \mathbb{R}.$$

The space of quaternions is denoted $\mathbb{H}$.

Remark 8

$\mathbb{H}$ admits the structure of a division ring. However, multiplication fails to be commutative, so $\mathbb{H}$ is not a field: if $q_1 = i$ and $q_2 = j$, $q_1 q_2 = ij \neq ji = q_2 q_1$.

Note that $\mathbb{H} \cong \mathbb{C}^2 \cong \mathbb{R}^4$ by the following isomorphisms. First rewrite a quaternion as

$$x + yi + zj + wk = (x + yi) + (zj + wk) = (x + yi) + (zj + wij) = (x + yi) + (z + wi)j.$$

Then the isomorphisms are given by the maps

$$\mathbb{H} \longrightarrow \mathbb{C}^2 \longrightarrow \mathbb{R}^4$$

$$(x + iy) + (z + iw)j \longmapsto (x + iy,\ z + iw) \longmapsto (x, y, z, w).$$

The quaternions can also be identified by a specific subgroup of the space of 2×2 complex matrices.

Definition 9
The Pauli matrices are a set of three 2×2 matrices $\{\sigma_1, \sigma_2, \sigma_3\} \subset \mathrm{Mat}_{2\times 2}(\mathbb{C})$ which are traceless, Hermitian, unitary, and involutive:

$$\sigma_1 := \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \qquad \sigma_2 := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \qquad \sigma_3 := \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

By sending $i \mapsto \sigma_1$, $j \mapsto \sigma_2$, and $k \mapsto \sigma_3$, any quaternion $q = x + yi + zj + wk$ can be rewritten as the linear combination

$$\begin{aligned} x\,\mathrm{Id} + y\sigma_1 + z\sigma_2 + w\sigma_3 &= \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} + \begin{pmatrix} yi & 0 \\ 0 & -yi \end{pmatrix} + \begin{pmatrix} 0 & z \\ -z & 0 \end{pmatrix} + \begin{pmatrix} 0 & wi \\ wi & 0 \end{pmatrix} \\ &= \begin{pmatrix} x + yi & z + wi \\ -z + wi & x - yi \end{pmatrix}. \end{aligned}$$

Now, notice that the $(2,2)$ entry is the complex conjugate of the $(1,1)$ entry, and the $(2,1)$ entry is the negative conjugate of the $(1,2)$ entry. This relationship is clearly from the chosen basis vectors, so it holds for the image of any quaternion:

$$q \longmapsto \begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix} \qquad \text{where} \qquad q = \alpha + \beta j.$$

By now, matrices of this form should be familiar. If we divide by the norm, we recover the SU(2) matrices!

Proposition 10
The Lie group $\mathrm{SU}(2) = \left\{ \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} \;\middle|\; \alpha, \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1 \right\}$ is isomorphic (as a ring) to the ring of unit quaternions.

Recall that SU(2) is the 2×2 complex matrices of determinant one. By identifying $\mathbb{C}^2$ with $\mathbb{R}^4$, an element of SU(2) determines a determinant one 4×4 matrix, i.e., a rotation in $\mathbb{R}^4$. Rotations in $\mathbb{R}^4$ are determined by a rotation of the unit sphere $S^3 \subset \mathbb{R}^4$, which is determined by the image of the $(1,0,0,0)$ vector, which is a point on S^3. This suggests the following claim.

Proposition 11
SU(2) is diffeomorphic to S^3.

Exercise 12
Prove this proposition.

All together, we can conclude that $\mathrm{SU}(2) \cong \{q \in \mathbb{H} \mid |q|^2 = 1\} \cong S^3$.

We can now ask: what is $\mathbb{HP}^1$? We now have the tools to understand why $\mathbb{HP}^1 \cong S^4$.

Recall that one construction of the projective spaces $\mathbb{R}P^1$ and $\mathbb{C}P^1$ are as the one point compactifications of $\mathbb{R}$ and $\mathbb{C}$, respectively; since $\mathbb{H} \cong \mathbb{R}^4$ and $\mathbb{R}^4 \cup \{\infty\} \cong S^4$, we can conclude that $\mathbb{HP}^1 \cong S^4$!

Alternatively, recall that $\mathbb{R}P^n$ is most commonly constructed as the quotient of S^n by the antipodal map. By viewing the antipodal map as the action of the "unit circle" $\{\pm 1\} \subset \mathbb{R}$, and further identifying $\{\pm 1\} \cong S^0$, $\mathbb{R}P^n$ can be seen as the base space of a fibration $p : S^n \to \mathbb{R}P^n$ with S^0 fibres. This construction is easily generalized to $\mathbb{C}P^n$ and then $\mathbb{HP}^n$: let V be a vector space over the ring $k = \mathbb{R}, \mathbb{C}$, or $\mathbb{H}$, and let $U(V^d)$ be the unit circle in V^d.

When $k = \mathbb{R}$, V^{n+1} is $n+1$ real dimensions, so $U(V_{\mathbb{R}}^{n+1}) \cong S^{n+1-1} = S^n$. The unit circle in $\mathbb{R}$ is $\{\pm 1\} \cong S^0$. This recovers the fibration

$$S^0 \to S^n \to \mathbb{R}P^n.$$

When $k = \mathbb{C}$, V^{n+1} is $2(n+1)$ real dimensions, so $U(V_{\mathbb{C}}^{n+1}) \cong S^{2(n+1)-1} = S^{2n+1}$. The unit circle in $\mathbb{C} \cong \mathbb{R}^2$ is $\{e^{i\theta}\} \cong S^1$, so the fibration becomes

$$S^1 \to S^{2n+1} \to \mathbb{C}P^n.$$

Note that when $n = 1$, this recovers the standard Hopf fibration $S^1 \to S^3 \to S^2$.

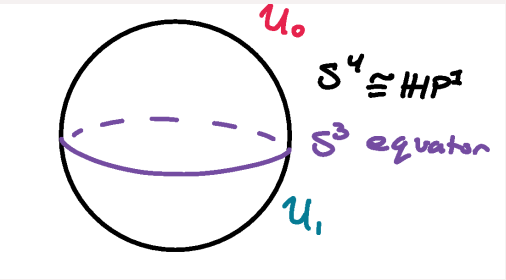
When $k \cong \mathbb{H}$, V^{n+1} is $4(n+1)$ real dimensions, so $U(V_{\mathbb{H}}^{n+1}) \cong S^{4(n+1)-1} = S^{4n+3}$. The unit circle in $\mathbb{H} \cong \mathbb{R}^4$ is S^3, thus the fibration is

$$S^3 \to S^{4n+3} \to \mathbb{HP}^n.$$

Instantons on S^4

Mirroring our previous discussion of instantons on $\mathbb{C}P^1$, we'll view $S^4 = \mathbb{HP}^1$. $\mathbb{HP}^1$ admits two charts $U_0, U_1 \cong \mathbb{R}^4 \cong \mathbb{H}$, with the coordinate change function given by $q \mapsto \frac{1}{q}$.

What are SU(2) bundles over S^4?



SU(2) bundles correspond to maps $S^3 \to \mathrm{SU}(2)$, where S^3 is the region on which we glue U_0 to U_1. Since $\mathrm{SU}(2) \cong S^3$, SU(2) bundles over S^4 are classified by maps $S^3 \to S^3$, of which there are $\mathbb{Z}$-many (degree).

In terms of coordinates, some possible gluing maps are $g(q) = q^n/|q|^n$.

The $k = 1$ bundle corresponds to $g(q) = q$.